# From Developer to CEO

## The Perfect Blend of Technology and Business



**Assoc. Prof. Dr. Hj. Mohamad Hafis Izran bin Ishak**

CEO, AR Mechatronics PLT  Universiti Teknologi Malaysia

# Your app works. Who will pay for it?

What problem is valuable enough for someone to fund the solution?



# Dr Hafis Izran



## Associate Professor, UTM

Control and Mechatronics Engineering Department (CMED)
Faculty of Electrical Engineering

## CEO, AR Mechatronics PLT

UTM spin-off company since 2019

Human adaptive mechatronics
Artificial intelligence, mobile apps and web development

# Engineering, industry and entrepreneurship

## 2002 / 2011

### Engineering foundation

UTM degree
PhD, Loughborough

## 2018–2019

### Industry exposure

CEO@Faculty Fellow
Alcon (M) Sdn. Bhd.

## 2019

### Company leadership

CEO, AR Mechatronics
from 28 May 2019

## 2024–2025

### Commercial recognition

INATEX recognition
EECS Industry Award



# Technology in different operating contexts

AR MECHATRONICS PLT
Simplify the difficulties.



A UTM spin-off since 2019

**MyREDS**

Community management

**Digital defect counter**

Industrial digitisation

**eBencana / e-Bantuan**

Disaster support systems

**GeoTrees collaborations**

Tree tagging and community projects

Selected examples from Dr Hafis's CV and supplied sharing material

# TIS in a working business

| TIS component | MyREDS example | Business effect |
|---|---|---|
| People | Residents and guards | Training and support |
| Hardware | Phones and gate devices | Installation and upkeep |
| Software | Apps and dashboard | Reliability and updates |
| Data | Access logs and reports | Useful records and privacy |
| Process | Register, verify, record | Consistent operations |

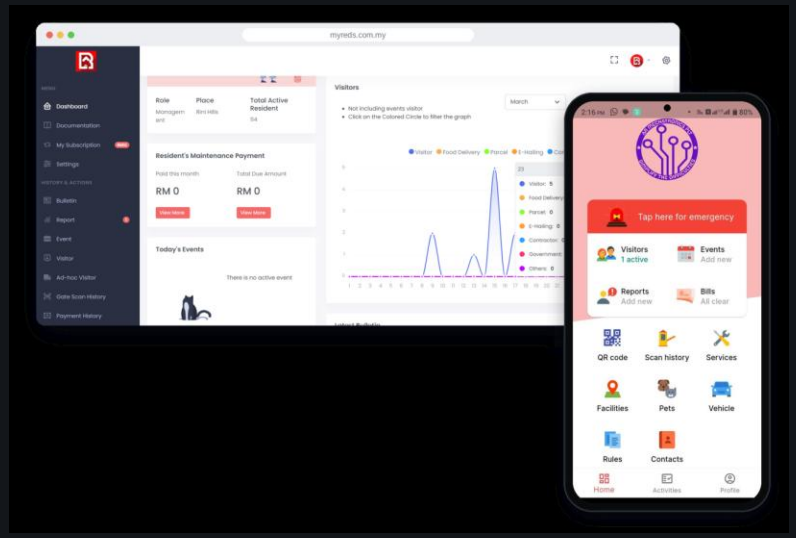


# A community system used every day

## MyREDS

Resident Easy
Decision System

Resident app
Guard app
Web admin dashboard





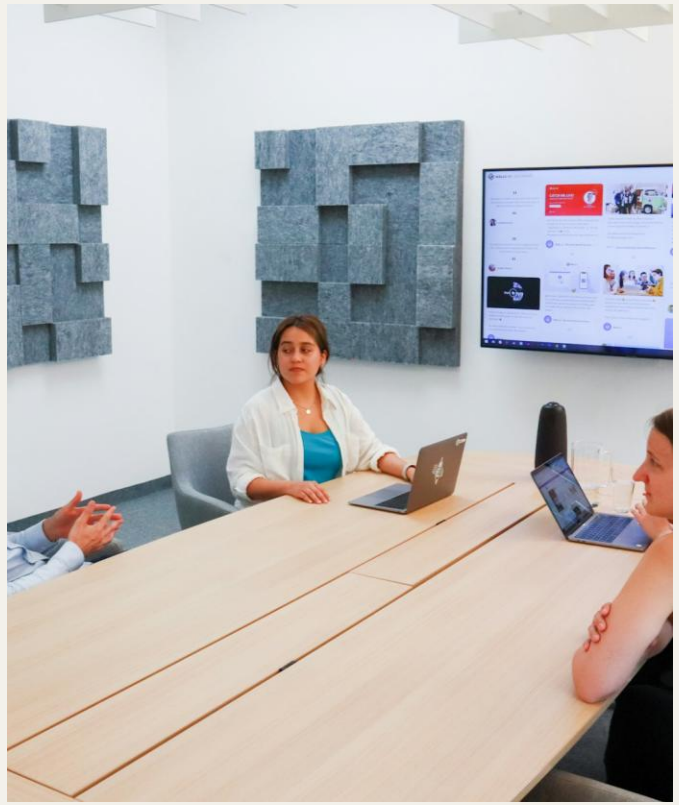

# One visitor, five handoffs

| | | |
|---|---|---|
| 01 | **Resident** | Creates an invitation |
| 02 | **Visitor** | Receives a QR pass or link |
| 03 | **Guard** | Verifies access |
| 04 | **Gate operation** | Allows, rejects or escalates |
| 05 | **Management** | Reviews the record |



# The user and the buyer may be different

| Person | Main need | Buying conversation |
|---|---|---|
| Resident | A quick, clear task | Adoption and feedback |
| Visitor | Clear entry instructions | First-time experience |
| Guard | A dependable decision | Training and operating fit |
| Management | Control and traceability | Budget and renewal |



Typical buying roles vary by community and contract

# The developer and the CEO

**DEVELOPER**

## Does the system work?

Correct behaviour
Performance and integration
Testing and maintainability

**CEO**

## Can we keep delivering it?

Customer value and payment
People and operating costs
Commitments after launch



# Problem discovery

## “What do you do today?”

## “Where does it become slow or difficult?”

Current workaround
Frequency of the problem
Consequence of failure



# The first useful release

## One complete visitor workflow

A minimum viable product lets a real user finish a meaningful task.



**IN THE FIRST TEST**

**Invitation and QR delivery**
**Verification and recovery**
**A usable access record**

Illustrative release scope, not a history of the first MyREDS launch

# Technology choices carry business costs

| Technical choice | Benefit | Cost or obligation |
|---|---|---|
| Cloud services | Shared access and updates | Hosting and availability |
| Hardware integration | Software connected to site operations | Installation and upkeep |
| Custom features | Fits a specific workflow | Development and support |
| Activity logging | Traceability and diagnosis | Access control and retention |



# Trust is part of the product

## Access

Who may see or change a record?

## Purpose

Why does this task need the data?

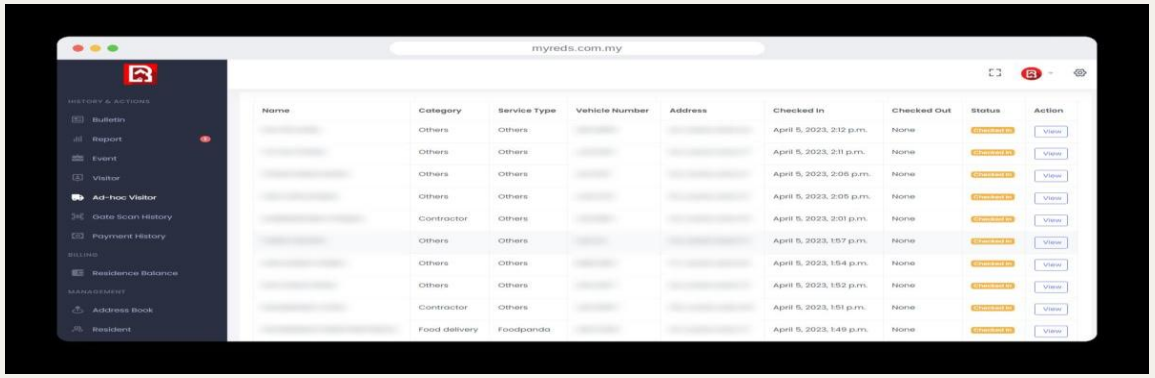
## Recovery

What happens when access fails?

## History

Can we explain the decision later?





# What does the customer pay for?

## Subscription

Software access and ongoing support

## Implementation

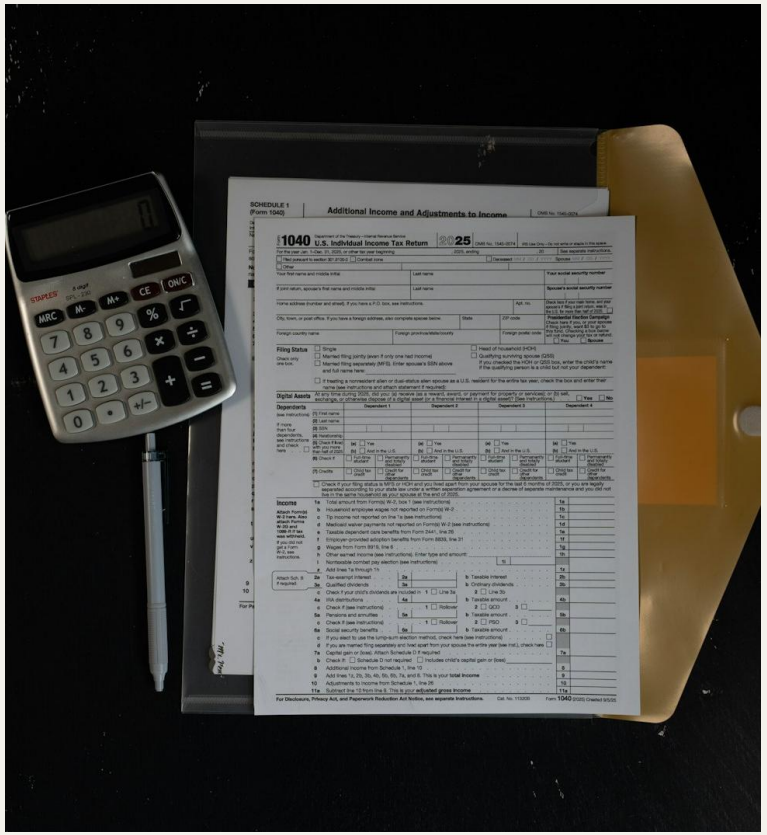
Configuration, onboarding and training

## Hardware

Devices, installation and maintenance

## Optional services

Integration or agreed development



# Revenue is only the starting point

| Monthly amount per community | RM |
|---|---|
| Subscription revenue | 300 |
| Direct service cost | 100 |
| Contribution to fixed costs | 200 |

**RM6,000 ÷ RM200 = 30**

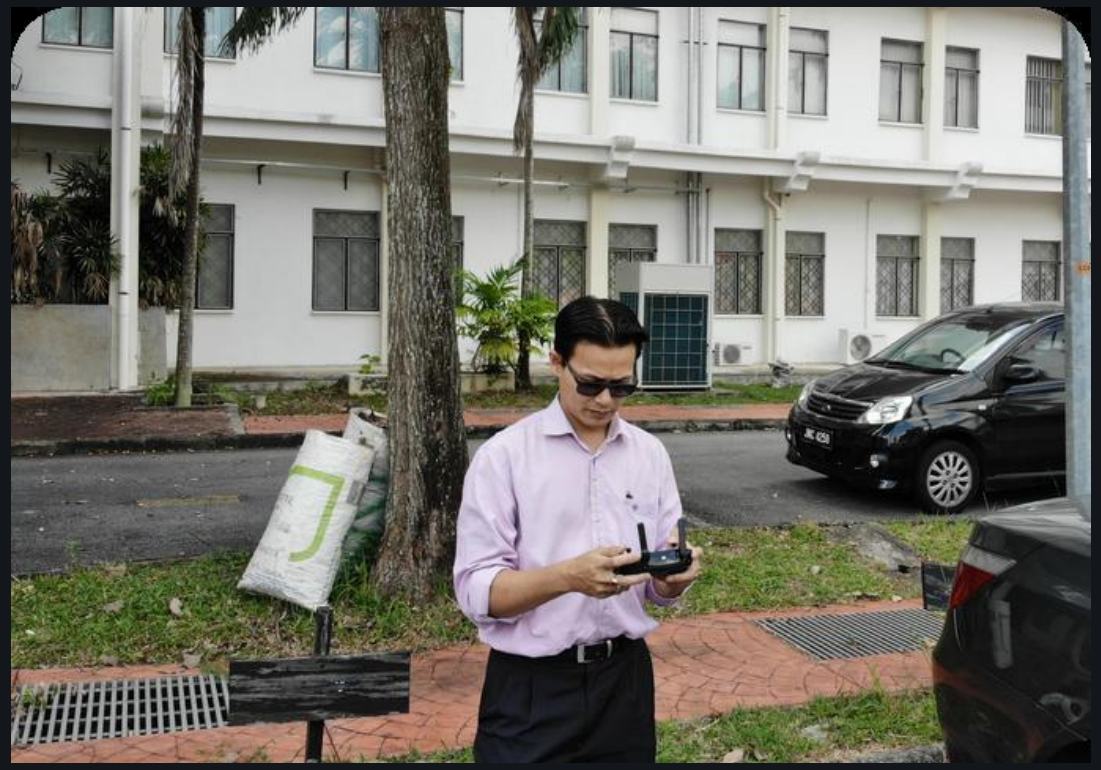
Hypothetical figures for learning. These are not MyREDS prices or company results.



**30** **communities**

To cover RM6,000 in monthly fixed costs

# A pilot tests the business promise



## Task success

Can users finish the workflow?

## Time and errors

Where does the process slow down?

## Support effort

How much help does the site need?

## Renewal intent

Would the buyer continue paying?

# A sale creates an operating commitment



**01** **Before signing**

Scope, payment terms and acceptance criteria

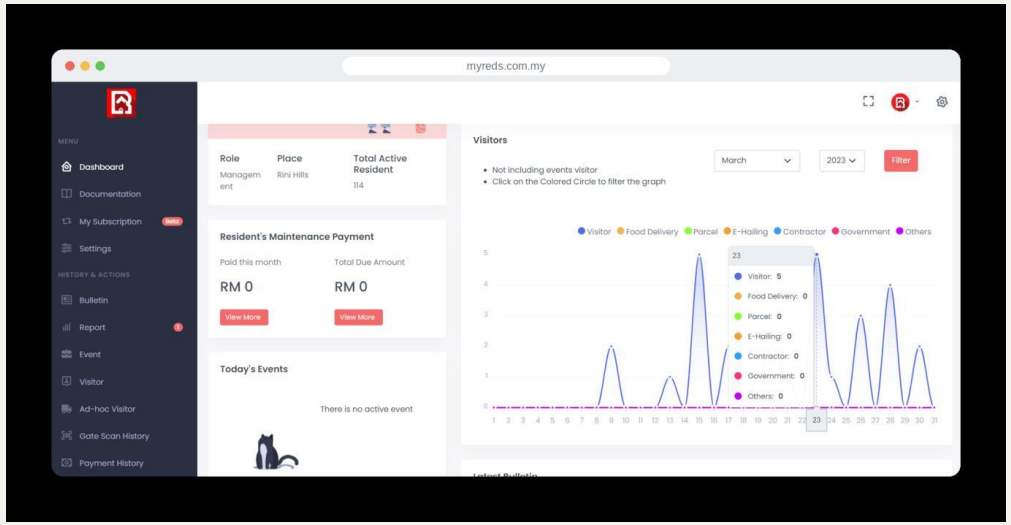
**02** **During rollout**

Site readiness, onboarding and a working fallback

**03** **After go-live**

Support, updates, incidents and billing or renewal

# Usage data can guide the next decision




Access errors: review the failure point.

Low usage: ask whether the task matters.

Heavy support: improve workflow or training.



# The job grows beyond your own code



## Ownership

A named person for each commitment

## Priorities

A shared view of what comes next

## People

Clear roles and useful feedback

## Cash discipline

Delivery costs and customer collections

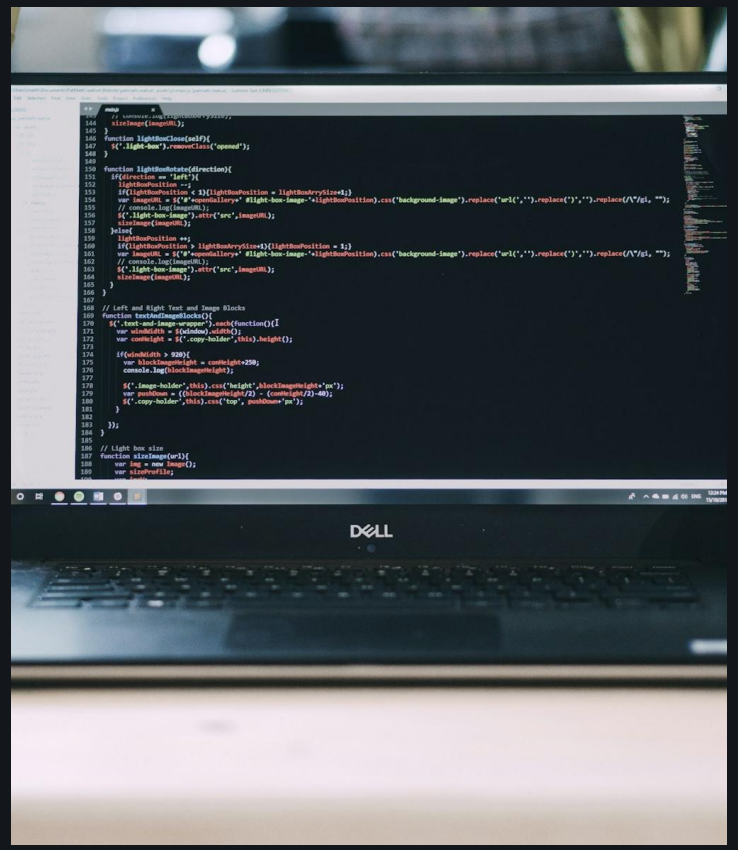
# AI assistance still needs human ownership

## Useful assistance

Draft a prototype
Explain an error
Suggest test cases

## Your responsibility

Architecture and integration
Validation and security
Customer commitments




A convincing demo still needs a dependable operating process

# Digital twins for community operations




## Physical site

Gates, roads and facilities

## Updated information

Access events or patrol locations

## Operator decision

Where should attention go now?

PROPOSED EXTENSION

**A view that updates with real site data**

# One product decision, two perspectives

| Decision | Technology perspective | Business perspective |
|---|---|---|
| New feature | Can we build and maintain it? | Who needs it enough to pay? |
| Availability | How will it recover? | What does disruption cost? |
| Integration | What must connect reliably? | Who installs and supports it? |
| Data collection | What supports the task? | What value justifies the effort? |



# Your five-minute product challenge

## A campus problem you can observe



Hostel visitors, facility bookings, parking or a student helpdesk

01 Who uses it, and who could pay?

02 What is difficult today?

03 What is the smallest complete workflow?

04 What costs money after launch?

05 What evidence would justify a pilot?

# A 30-day student roadmap

WEEK 1

## Observe

Speak with five users
and map the current task.

**Output: problem brief**

WEEK 2

## Prototype

Build one complete
workflow to test.

**Output: usable prototype**

WEEK 3

## Test

Watch users attempt
the task and record issues.

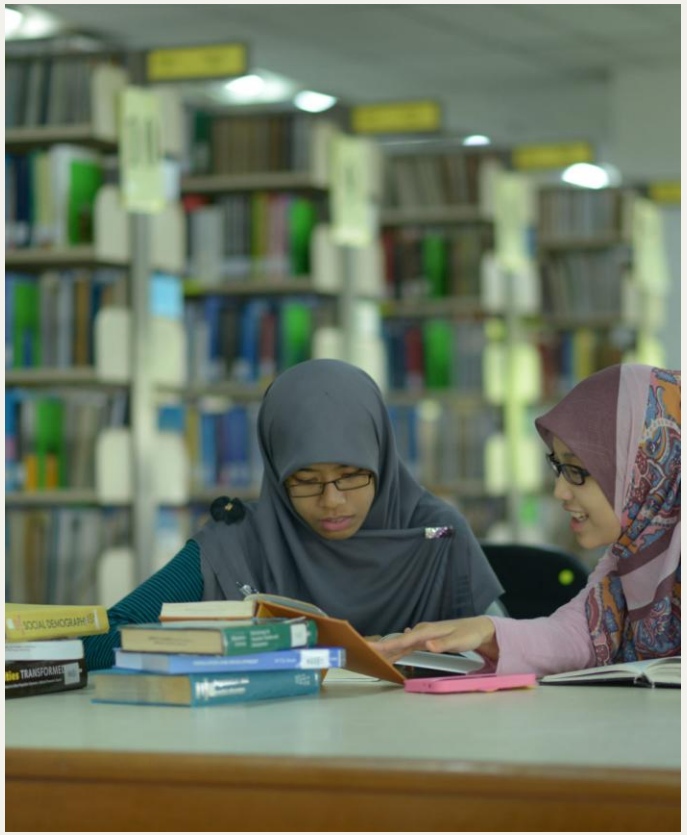
**Output: evidence log**

WEEK 4

## Explain the value

Revise the product.
Estimate the costs.

**Output: demo and pitch**



# Five lessons for your next project

01 Strong code needs a clear problem

02 Users and buyers may be different

03 Delivery continues after launch

04 Costs shape product choices

05 Evidence improves the next version



# What would you build for someone who would actually use it?

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